

BACHELORS ISLAND, MB, Canada

WMO: 711430

Lat: **51.750N**Lon: **99.900W**Elev: **256**StdP: **98.29**Time Zone: **-6.00 (NAC)**Period: **94-10**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.8	-30.1	-36.0	0.1	-32.7	-33.2	0.2	-30.0	14.2	-7.9	12.4	-13.6	2.9	320	0.749

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	26.6	21.7	25.3	21.1	24.0	20.3	22.7	25.6	21.7	24.5	20.7	23.5	5.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.7	16.9	25.2	20.6	15.8	24.0	19.5	14.7	23.0	68.7	25.6	64.7	24.5	61.0	23.7	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.7	10.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-18.3	-13.7	-6.9	0.9	9.0	16.3	19.9	19.0	13.6	6.1	-3.9	-12.5
	DBStd	14.06	7.56	7.67	7.91	5.02	4.87	3.63	2.64	3.25	4.14	3.36	5.45	7.72
	HDD10.0	3673	877	663	525	274	78	2	0	0	14	126	416	697
	HDD18.3	5903	1136	897	784	524	289	78	15	31	150	379	666	955
	CDD10.0	954	0	0	0	1	48	193	306	280	122	5	0	0
	CDD18.3	142	0	0	0	0	1	19	62	52	8	0	0	0
	CDH23.3	428	0	0	0	0	6	49	191	169	12	0	0	0
	CDH26.7	44	0	0	0	0	1	6	16	21	1	0	0	0
Wind	WSAvg	5.1	4.3	4.5	4.7	4.8	5.4	5.3	5.1	5.4	5.9	5.8	5.4	4.8
Precipitation	PrecAvg	520	28	14	31	28	52	83	77	61	52	36	26	30
	PrecMax	610	75	30	61	52	115	128	131	110	101	62	91	56
	PrecMin	436	6	1	9	5	16	31	30	29	9	10	6	9
	PrecStd	41	16	8	17	14	26	30	29	23	27	16	21	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.4	2.1	8.9	14.9	23.5	27.2	28.4	28.7	25.2	15.4	8.0	4.6
		MCWB	2.5	1.1	4.6	8.7	15.4	21.1	22.9	22.6	19.1	11.9	4.8	2.3
	2%	DB	-0.8	1.1	5.2	11.7	19.7	24.6	26.6	26.7	22.7	13.1	5.6	2.0
		MCWB	-1.4	0.4	2.9	7.6	15.0	20.5	21.8	22.0	18.9	9.6	3.2	0.8
	5%	DB	-4.4	0.1	3.8	8.9	17.5	22.9	25.4	25.1	21.3	11.7	4.0	-0.8
		MCWB	-5.0	-0.5	2.3	6.0	13.8	19.2	21.2	21.3	18.0	9.1	2.3	-1.6
	10%	DB	-7.0	-2.3	2.5	6.7	15.8	21.5	24.2	23.8	19.6	10.5	2.8	-2.5
		MCWB	-7.5	-2.8	1.3	4.4	12.5	18.3	20.5	20.4	16.5	8.6	1.6	-3.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.2	1.2	5.0	9.8	17.1	23.0	24.2	24.0	20.2	12.3	5.0	2.4
		MCDB	4.0	1.6	7.9	13.4	21.4	25.1	27.0	27.6	23.3	14.8	7.4	4.1
	2%	WB	-1.3	0.5	3.3	8.0	15.6	21.1	22.7	22.9	19.4	10.6	3.5	0.7
		MCDB	-0.8	1.0	5.0	10.8	18.8	23.6	25.6	25.8	22.4	12.2	5.3	2.0
	5%	WB	-4.9	-0.6	2.2	6.2	14.2	19.8	21.7	21.7	18.3	9.8	2.5	-1.5
		MCDB	-4.4	0.0	3.5	9.1	17.3	22.4	24.7	24.4	21.1	11.3	3.9	-0.7
	10%	WB	-7.4	-2.8	1.3	4.4	12.6	18.6	20.8	20.7	16.8	8.9	1.6	-3.1
		MCDB	-7.0	-2.3	2.4	6.5	15.4	21.1	23.7	23.4	19.4	10.5	2.8	-2.5
Mean Daily Temperature Range	5% DB	MDBR	8.4	9.8	9.8	8.3	6.5	5.9	6.3	5.9	5.6	4.6	5.4	7.1
		MCDBR	9.1	8.1	10.0	10.6	9.2	8.4	8.0	7.6	7.8	6.8	6.3	9.0
		MCWBR	8.7	7.7	8.1	7.0	5.5	5.5	5.4	4.8	5.1	4.7	4.9	8.3
	5% WB	MCDBR	8.7	7.9	9.4	10.5	7.6	7.3	7.5	7.2	7.1	6.4	6.2	8.9
		MCWBR	8.3	7.5	7.8	7.0	4.9	5.3	5.3	5.0	5.0	4.7	5.1	8.3
Clear-Sky Solar Irradiance	taub		0.266	0.273	0.290	0.352	0.361	0.382	0.385	0.375	0.342	0.296	0.265	0.249
	taud		2.237	2.261	2.364	2.354	2.393	2.361	2.385	2.405	2.483	2.570	2.433	2.282
	Ebn at Noon		743	854	913	886	892	872	863	853	846	835	761	709
	Edh at Noon		66	86	96	110	112	117	112	104	86	65	55	53
All-Sky Solar Radiation	RadAvg		1.17	2.33	3.76	5.01	5.54	5.72	6.02	4.94	3.54	2.01	1.05	0.83
	RadStd		0.10	0.15	0.26	0.37	0.49	0.42	0.27	0.39	0.33	0.26	0.14	0.08

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (12 neighbors)	N/A	N/A	N/A	N/A	-0.45	-0.56	N/A	N/A	N/A		

Nomenclature: See separate page